



CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 20: Cylinder, Cone and Sphere (Surface Area and Volume)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sphere and hemisphere

1. The volumes of two spheres are in the ratio 27 : 64. Find the ratio of their radii. (a) 3 : 4 (b) 4 : 3 (c) 9 : 16 (d) 16 : 9 **ICSE 2025, 1 mark**

SOLUTION

Volume is proportional to r^3 : $\frac{r_1^3}{r_2^3} = \frac{27}{64}$, so $\frac{r_1}{r_2} = \frac{3}{4}$.

Answer: (a)

2 Melting and recasting

2. A hollow sphere has an external diameter of 10 cm and an internal diameter of 6 cm. It is melted and made into a solid cone of height 8 cm. Find the radius of the cone. **ICSE 2025, 3 marks**

SOLUTION

$$\text{Volume of metal} = \frac{4}{3}\pi(R^3 - r^3) = \frac{4}{3}\pi(5^3 - 3^3) = \frac{4}{3}\pi \times 98$$

Volume of the cone = $\frac{1}{3}\pi x^2 \times 8$, where x is its radius.

$$\frac{8}{3}\pi x^2 = \frac{4}{3}\pi \times 98, \text{ so } x^2 = 49.$$

Answer: 7 cm



3 Combination of solids

3. A solid wooden capsule is a cylinder with a hemisphere on each end. Its radius is 3.5 cm and the cylindrical part is 14 cm long. The capsule is cut into three separate solids: the cylinder and the two hemispheres. Find the sum of the total surface areas of the three pieces. Take $\pi = \frac{22}{7}$.

ICSE 2025, 4 marks

SOLUTION

Each piece is now a separate solid, so its flat faces count as well.

$$\text{Cylinder: } 2\pi r(h + r) = 2 \times \frac{22}{7} \times 3.5 \times (14 + 3.5) = 22 \times 17.5 = 385 \text{ cm}^2$$

$$\text{One solid hemisphere: } 3\pi r^2 = 3 \times \frac{22}{7} \times 3.5 \times 3.5 = 115.5 \text{ cm}^2. \text{ Two hemispheres: } 231 \text{ cm}^2.$$

$$\text{Sum} = 385 + 231$$

Answer: 616 cm²

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